Lecture 28

Airy functions, Stokes lines and WKB

Objectives. *The Airy function connects exponential decay on one real side to oscillation on the other. We compute the two oscillatory saddle contributions and derive the formal WKB amplitudes. A simple turning point reduces locally to the Airy model. The stationary-phase input and the general matching argument retain their sketch qualifications.*

By the end of this lecture you will be able to:

- Compute the phases and constants in the sketched negative-Airy leading term.
- Distinguish equal-phase and equal-magnitude rays under an explicit naming convention.
- Derive the WKB transport equation and its normalised differential residual.
- Explain the sketched simple-turning-point connection and its overlap range.

28.1 The two oscillatory Airy saddles

Recall from Proposition 14.5 that the entire Airy function satisfies $\mathrm{Ai}''(z) = z\,\mathrm{Ai}(z)$ and, for real arguments, has the improper oscillatory representation

$$\mathrm{Ai}(-x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{\mathrm{i}(v^3/3 - xv)}\,\mathrm{d}v, \qquad x > 0.$$

The one-sided improper integrals converge by integration by parts at infinity. Set $v = \sqrt{x}\,u$. The phase becomes $x^{3/2}\varphi(u)$ with $\varphi(u) = u^3/3 - u$. Its stationary points are $u = -1$ and $u = 1$. Their values and second derivatives are

$$\varphi(-1) = \frac{2}{3}, \quad \varphi''(-1) = -2, \qquad \varphi(1) = -\frac{2}{3}, \quad \varphi''(1) = 2.$$

Proposition 28.1 (Negative Airy leading term · *sketched*)**.** *As $x \to +\infty$,*

$$\mathrm{Ai}(-x) = \frac{x^{-1/4}}{\sqrt{\pi}}\left(\sin\left(\frac{2}{3}x^{3/2} + \frac{\pi}{4}\right) + o(1)\right). \tag{28.1}$$

The error is additive, of size $o(x^{-1/4})$.

Proof sketch. Choose smooth compactly supported cutoffs χ_- and χ_+ equal to one near -1 and 1, respectively, with disjoint supports. Apply the sketched stationary-phase theorem, Theorem 14.4, to each localised integral. Write $X = 2x^{3/2}/3$ for this calculation. After multiplication by $\sqrt{x}/(2\pi)$, the two contributions are

$$\frac{x^{-1/4}}{2\sqrt{\pi}}\left(e^{\mathrm{i}(X-\pi/4)} + e^{\mathrm{i}(-X+\pi/4)}\right) = \frac{x^{-1/4}}{\sqrt{\pi}}\sin(X + \pi/4).$$

Each localised remainder is $o(x^{-1/4})$ by the inherited sketch. On the complement put $b = 1 - \chi_+ - \chi_-$. The quotient b/φ' is smooth, tends to zero at both infinities, and its derivative is integrable. Integration by parts gives

$$\int_{\mathbb{R}} b(u) e^{\mathrm{i}x^{3/2}\varphi(u)}\,\mathrm{d}u = -\frac{1}{\mathrm{i}x^{3/2}} \int_{\mathbb{R}} \left(\frac{b}{\varphi'}\right)' e^{\mathrm{i}x^{3/2}\varphi(u)}\,\mathrm{d}u = O(x^{-3/2}).$$

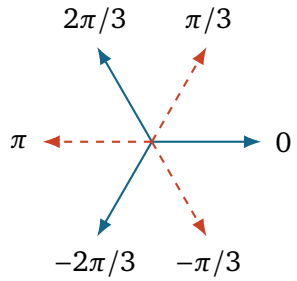
Thus the prefactor makes this outer contribution $O(x^{-1})$, smaller than $x^{-1/4}$. Adding the two saddle terms proves the stated leading calculation. Its tag records the inherited stationary-phase sketch. ∎

A ratio to the displayed sine is not controlled near its zeros. The additive formulation describes the oscillation without dividing by a quantity that vanishes infinitely often.

28.2 Equal-phase and equal-magnitude rays

On the slit $-\pi < \arg z < \pi$, define $\zeta(z) = 2z^{3/2}/3$ using the principal power. The formal Airy exponentials are $e^{-\zeta}$ and e^{ζ}. We call $\operatorname{Im}\zeta = 0$ the *Stokes rays*, meaning equal imaginary exponents, and $\operatorname{Re}\zeta = 0$ the *anti-Stokes rays*, meaning equal magnitudes. Some sources reverse these names, so the equations specify our convention.

Writing $z = re^{\mathrm{i}\alpha}$ gives $\zeta = (2/3)r^{3/2}e^{3\mathrm{i}\alpha/2}$. Consequently the Stokes rays have arguments 0 and $\pm 2\pi/3$. The anti-Stokes rays have arguments $\pm\pi/3$ and the two boundary values on the negative axis. The latter represent the same geometric ray. The rays are shown below.



The solid rays have equal imaginary exponents. The dashed rays have equal exponential magnitudes. The negative-axis label denotes boundary values of the chosen power, not a branch of the entire Airy function.

The *Stokes phenomenon* is a change in a sectorial asymptotic description when a subdominant exponential contribution becomes relevant. A *subdominant* exponential has smaller modulus than its partner in that sector. The exact function Ai has no jump across any ray: it is entire. The ray calculation alone determines neither a connection multiplier nor a full sectorial asymptotic theorem. Those global results are cited to Ablowitz–Fokas, §6.6; none is needed for the real matching below.

28.3 The formal WKB balance

Let $\varepsilon > 0$ be small and let $Q(t)$ be a smooth real function on an interval. Consider the equation

$$\varepsilon^2 y''(t) = Q(t)y(t). \tag{28.2}$$

The variable t is the independent coordinate, not the Airy argument z. On a subinterval where $Q > 0$, insert $y = Be^{S_0/\varepsilon}$, with nonzero B independent of ε at leading order. Direct differentiation gives

$$\frac{\varepsilon^2 y''}{y} = (S_0')^2 + \varepsilon\left(2\frac{B'}{B}S_0' + S_0''\right) + \varepsilon^2\frac{B''}{B}.$$

The order-one equation $(S_0')^2 = Q$ is the *eikonal equation*. For either choice $S_0' = \pm\sqrt{Q}$, the order-ε equation is $2B'S_0' + BS_0'' = 0$, called the *transport equation*. Since $S_0''/S_0' = Q'/(2Q)$, it gives $B'/B = -Q'/(4Q)$. The resulting *WKB leading approximants* are

$$y_\pm(t) = Q(t)^{-1/4}\exp\left(\pm\frac{1}{\varepsilon}\int^t \sqrt{Q(s)}\,\mathrm{d}s\right). \tag{28.3}$$

For $Q < 0$, replace the amplitude by $(-Q)^{-1/4}$ and the exponent by $\pm\mathrm{i}\varepsilon^{-1}\int^t\sqrt{-Q(s)}\,\mathrm{d}s$. In this case the real and imaginary parts of the branches are oscillatory cosine and sine approximants.

Proposition 28.2 (Differential residual)**.** *For either WKB branch on an interval where $Q \neq 0$, the exact quotient is*

$$\frac{\varepsilon^2 y_\pm'' - Qy_\pm}{Qy_\pm} = \varepsilon^2\frac{5(Q')^2 - 4QQ''}{16Q^3}. \tag{28.4}$$

Proof. The eikonal and transport terms cancel in the differentiated formula. Logarithmic differentiation of $B = |Q|^{-1/4}$ gives

$$\frac{B''}{B} = \left(-\frac{Q'}{4Q}\right)' + \frac{(Q')^2}{16Q^2} = \frac{5(Q')^2 - 4QQ''}{16Q^2}.$$

The remaining quotient is $\varepsilon^2 B''/(QB)$, as claimed. The same cancellation holds with imaginary eikonal derivative for $Q < 0$. ∎

The numerator is the *differential residual*, the error after inserting an approximant in the differential equation. A small residual is not by itself a theorem bounding the difference from an exact solution. In particular, it does not control a solution without initial or boundary information.

28.4 A simple turning point and its Airy model

A *simple turning point* is a zero of Q at which $Q' \neq 0$. Translate it to $t = 0$ and suppose $Q'(0) = a > 0$. Assume Q is smooth near zero. Then $Q(t) = at + O(t^2)$, with $Q > 0$ to the right and $Q < 0$ to the left on a small fixed neighbourhood. The square roots used below on the positive and negative sides are positive real roots. Define the actions by

$$S(t) = \int_0^t \sqrt{Q(s)}\,\mathrm{d}s \quad (t > 0), \qquad T(t) = \int_t^0 \sqrt{-Q(s)}\,\mathrm{d}s \quad (t < 0).$$

The decaying right-hand approximant is $CQ^{-1/4}e^{-S/\varepsilon}$.

First take the exact linear model $Q(t) = at$. The scale $u = a^{1/3}t/\varepsilon^{2/3}$ converts the equation to $\mathrm{d}^2y/\mathrm{d}u^2 = uy$. Thus $D\,\mathrm{Ai}(u)$ is an exact solution. The positive-Airy estimate, Proposition 14.6, matches the right-hand WKB coefficient C when

$$D = 2\sqrt{\pi}\,C(a\varepsilon)^{-1/6}.$$

Indeed its positive-side amplitude is $Da^{-1/12}\varepsilon^{1/6}t^{-1/4}/(2\sqrt{\pi}) = C(at)^{-1/4}$. The negative-Airy estimate, Proposition 28.1, has twice this amplitude, with phase $T/\varepsilon + \pi/4$. The actions are exactly $2\sqrt{a}\,t^{3/2}/3$ on the right and $2\sqrt{a}\,(-t)^{3/2}/3$ on the left. This proves the matching constants conditional on the two sketched Airy estimates.

For the linear coefficient $Q(t) = t$, the next figure compares one exact Airy solution with its two outer descriptions. The smooth exact curve remains finite at the turning point; the WKB amplitude has no finite value there. The calculation following the figure fixes the normalisation.

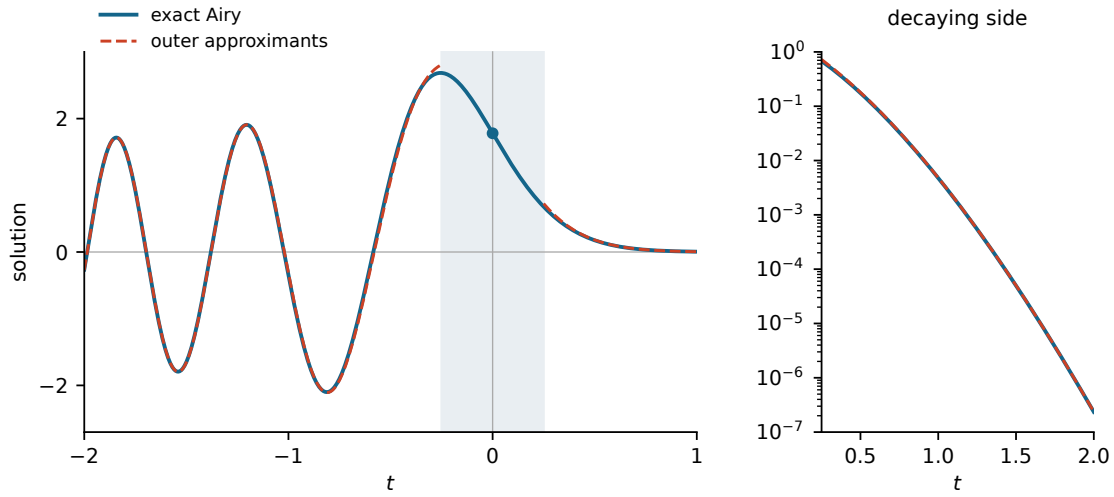


Figure 28.1: The exact linear model at $\varepsilon = 1/8$, normalised to right-hand WKB coefficient one. Dashed outer approximants are drawn only for $|t| \geq \varepsilon^{2/3}$; the shaded strip marks the central scale. The separate logarithmic plot compares the decaying tails.

Select the coefficient in the exact family

$$y(t) = D\,\mathrm{Ai}(t/\varepsilon^{2/3}), \qquad \varepsilon > 0,$$

by requiring the decaying normalisation

$$\lim_{t\to+\infty} t^{1/4}e^{2t^{3/2}/(3\varepsilon)}y(t) = 1.$$

The positive-Airy formula makes the limit $D\varepsilon^{1/6}/(2\sqrt{\pi})$. Thus the selected solution and its left-hand description are

$$D = 2\sqrt{\pi}\,\varepsilon^{-1/6},$$
$$y(t) = 2(-t)^{-1/4}\left(\sin\left(\frac{2(-t)^{3/2}}{3\varepsilon} + \frac{\pi}{4}\right) + o(1)\right) \qquad (t \to -\infty).$$

Here the parameter is fixed while the Airy argument tends to infinity. Equivalently, the same leading formula holds at a fixed negative point as $\varepsilon \downarrow 0$. In a real sine-and-cosine basis the coefficient selection reads

$$2\sin\left(v + \frac{\pi}{4}\right) = \sqrt{2}\sin v + \sqrt{2}\cos v.$$

Both left-hand coefficients are therefore $\sqrt{2}$ when the right-hand coefficient is one. They cannot be chosen independently for this selected Airy family. At the turning point the exact solution instead has the finite initial data

$$y(0) = D\,\mathrm{Ai}(0), \qquad y'(0) = D\varepsilon^{-2/3}\,\mathrm{Ai}'(0).$$

These data determine the analytic solution locally by Theorem 23.1. Any two entire continuations of this local solution agree by the identity theorem, Theorem 10.6. The coefficient matching uses the sketched Airy estimates; the scaled ODE and the initial-data identities are exact.

Proposition 28.3 (Simple-turning-point matching · *sketched*)**.** *The usual leading matching construction for the smooth simple zero above connects the decaying and oscillatory approximants by*

$$CQ(t)^{-1/4}e^{-S(t)/\varepsilon} \quad \longleftrightarrow \quad 2C(-Q(t))^{-1/4}\sin\left(\frac{T(t)}{\varepsilon}+\frac{\pi}{4}\right). \tag{28.5}$$

The two expressions apply on $t > 0$ and $t < 0$, respectively. The arrow identifies their leading matching coefficients, not equality at the same point or a proved uniform error bound through zero.

Matching sketch. Outside zero, use the WKB amplitudes and actions. For $Q(t) = at + O(t^2)$, the action differs from the linear action by $O(|t|^{5/2})$. This follows by expanding $\sqrt{|Q(t)|} = \sqrt{a|t|}(1 + O(|t|))$ and integrating. The Airy argument must be large in the matching region, requiring $|t| \gg \varepsilon^{2/3}$. The difference of the actions divided by ε must tend to zero, requiring $|t| \ll \varepsilon^{2/5}$. These requirements have the nonempty overlap

$$\varepsilon^{2/3} \ll |t| \ll \varepsilon^{2/5}, \qquad \varepsilon \downarrow 0.$$

Here $u \ll v$ means $u/v \to 0$ along the comparison being made. In this overlap the linear Airy calculation gives the factor 2 and phase $\pi/4$ for the leading approximants on each side. Controlling exact solutions for a general Q needs an additional turning-point error theorem. This argument supplies the matching mechanism and constants, with the sketch qualification, rather than such a theorem. ∎

The overlap is an asymptotic range, rather than an interval of fixed width. The separate scale diagram shows its two requirements.

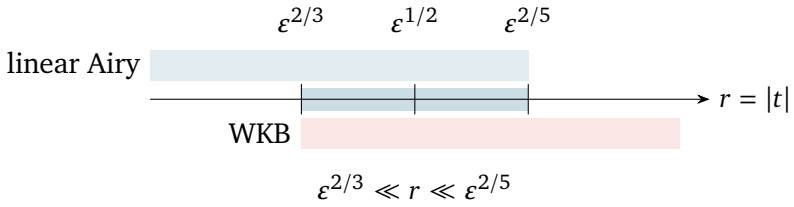


Figure 28.2: Schematic scale ordering for a general smooth simple zero as $\varepsilon \downarrow 0$. The upper and lower bands show the linear-Airy phase comparison and WKB regimes. Their strict overlap contains the choice $r = \varepsilon^{1/2}$; the marked endpoints are not included.

For a concrete path through this range, choose $r = |t| = \varepsilon^{1/2}$. Then

$$\frac{r}{\varepsilon^{2/3}} = \varepsilon^{-1/6} \to \infty, \qquad \frac{r}{\varepsilon^{2/5}} = \varepsilon^{1/10} \to 0.$$

The residual scale and the phase-comparison scale both tend to zero:

$$\frac{\varepsilon^2}{r^3} = \varepsilon^{1/2}, \qquad \frac{r^{5/2}}{\varepsilon} = \varepsilon^{1/4}.$$

This verifies compatibility of the two scale requirements. It still does not turn the general matching sketch into an exact-solution error theorem.

The oscillatory side always requires an additive error statement near zeros of its sine. Even in the exact linear model, the formula there is a leading sine plus $o(1)$ after division by its envelope, not a uniform relative error after division by the sine itself.

28.5 A residual and its range

For the linear model $Q(t) = at$, the normalised residual is $5\varepsilon^2/(16at^3)$. Its modulus is small when $|t| \gg \varepsilon^{2/3}$. It diverges at zero, where the Airy model remains regular. This explains why the WKB amplitudes alone cannot cover the turning point.

For a second example, take $Q(t) = t + t^2$ on $|t| < 1/2$. The origin is its only turning point in that interval and $Q'(0) = 1$. The numerator $5(Q')^2 - 4QQ''$ is $5 + 12t + 12t^2$, so

$$\frac{\varepsilon^2 y_\pm'' - Q y_\pm}{Q y_\pm} = \frac{\varepsilon^2(5 + 12t + 12t^2)}{16(t + t^2)^3}.$$

On every closed subinterval avoiding zero this is $O(\varepsilon^2)$. Near zero its leading magnitude is proportional to $\varepsilon^2/|t|^3$. The exact Airy scaling gives the appropriate local coordinate, but this residual calculation does not prove a uniform approximation for that nonlinear coefficient.

The final Airy problem combines the two parts of the course. The ODE gives an exact analytic model. Contour deformation and stationary phase give its asymptotic descriptions. Estimates identify the regimes where those descriptions can be compared. Each step answers a different question, so the exact equation alone does not prove an approximation error bound.

The same distinction recurs earlier. An integral limit needs a majorant or another stated convergence argument. A Fourier expansion needs its mode of convergence. An analytic continuation needs its domain and branch. A spectral or geometric conclusion retains the qualification of any cited input. Keeping the object, hypotheses and error statement together makes these methods usable in a new problem.

Summary.

- Two sketched stationary-phase contributions give an additive negative-Airy leading sine with envelope $\pi^{-1/2}x^{-1/4}$.
- Our Stokes convention is $\operatorname{Im}\zeta = 0$; anti-Stokes means $\operatorname{Re}\zeta = 0$. The exact entire Airy function does not jump across these rays.
- Eikonal and transport balance give WKB leading amplitudes. Their displayed normalised differential residual is exact away from zeros of Q.
- The sketched simple-zero connection has amplitude factor 2 and phase $\pi/4$. The linear-action matching overlap is $\varepsilon^{2/3} \ll |t| \ll \varepsilon^{2/5}$.

The analytic model and its asymptotic approximants have distinct roles: regularity of the model does not replace an approximation error theorem.

References

E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton Lectures II, 2003, Appendix A, §3, pp. 328–331, for Airy oscillatory normalisation. M. J. Ablowitz and A. S. Fokas, *Complex Variables: Introduction and Applications*, second edition, Cambridge, 2003, §6.6, pp. 488–494, for sectorial asymptotics and the Stokes convention. C. M. Bender and S. A. Orszag, *Advanced Mathematical Methods for Scientists and Engineers*, McGraw–Hill, 1978, §10.4, pp. 504–510, especially formulas (10.4.10)–(10.4.13), for the genuine overlap and simple-turning-point matching. Their coordinate x is t here.